

From Admissibility to Quantum Structure: Phase Coherence and Correlations from Non-Injective Projection

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Abstract

We establish that quantum-mechanical phase coherence is not an independent postulate but a structural consequence of projective admissibility within the Cosmochrony framework. The central result (Theorem 2.7) shows that any admissible transition preserves the Born–Infeld indiscernibility of conjugate Weil blocks ρ_c and ρ_{q-c} on the Heisenberg Cayley graph $\text{Cay}(\text{Heis}_3(\mathbb{Z}/q\mathbb{Z}), S_q)$, and thereby maintains the metaplectic phase coherence of the admissible fibre. The proof rests on three independently established results: the minimal parity fibre structure of the non-injective projection (O18), the dynamical identity of conjugate Weil blocks (O17), and the discrete admissibility filter of projection locking (O22). A constructive separating functional $R_n^{(c)}$ is exhibited and proved BI-admissible. Numerical validation for $q = 29$ across four conjugate pairs confirms that $\text{rank } W_n^{(c)} = 0$ exactly throughout the admissible regime, establishing structural indistinguishability of conjugate blocks as a representation-theoretic fact rather than a dynamical coincidence. As a structural consequence, the admissible fibre carries complex amplitude structure with interference, constituting the natural precursor of an effective Hilbert sector — without quantum mechanics as a postulate. The singlet correlator $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$ is then derived from admissibility, the parity involution (O18), and the isotropic $\text{SU}(2)$ structure (O23), without invoking any quantum postulate. As a consequence, the Tsirelson bound $|S_{\text{CHSH}}| \leq 2\sqrt{2}$ follows unconditionally from the geometry of $\text{Im } \mathbb{H}$ under admissible projection. Finally, the Born probability rule $P(a = +1 | \hat{a}) = |\langle +\hat{a} | \psi \rangle|^2$ is derived as the unique probability assignment compatible with the coherent fibre structure, the $\text{SU}(2)$ observable algebra, and the two-point correlator law. No quantum postulate is invoked at any step of the derivation.

Keywords. Non-injective projection; Quantum foundations; Emergence; Phase coherence; Born–Infeld structure; Weil representation; $SU(2)$ symmetry; Fiber structure; Admissibility; Gram–Schmidt dynamics

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1 Introduction

The structural origin of quantum-mechanical phase coherence is one of the deepest open questions in the foundations of physics. Standard formulations of quantum mechanics postulate the Hilbert space structure, complex amplitudes, and the superposition principle as axioms. The present paper shows that, within the projective admissibility framework of Cosmochrony [9], phase coherence is not postulated but derived: it is a necessary consequence of the admissibility constraint that governs which transitions between observable states are physically realised.

Context and position in the programme

The Cosmochrony framework [9] derives physical structure from a single primitive: the non-injective projection between successive observable states. Non-injectivity is not a modelling choice but a structural necessity — any projection compatible with genuine emergence must be non-injective, as established in [10]. The Born–Infeld structure is selected as the unique admissible completion of the effective action compatible with bounded projective flux [11].

Within this framework, the spectral admissibility sub-programme (O-series) has established a concrete chain of results on the Heisenberg Cayley graph $\text{Cay}(\text{Heis}_3(\mathbb{Z}/q\mathbb{Z}), S_q)$: the minimal parity fibre $\{c, q - c\}$ forced by Born–Infeld evenness (O18), the discrete admissibility filter of projection locking (O22), and the derivation of exactly three stable neutral directions from quaternionic minimality (O23).

The present paper establishes the bridge between this spectral programme and quantum mechanics. It answers the question: why does the admissible fibre carry complex amplitudes and support interference, rather than being a mere set of classical alternatives?

The present paper sits at the intersection of three lines of work: the structural failure of Bell factorisation from non-injective projection [12]; the spectral admissibility sub-programme (O-series), which establishes concrete results on the Heisenberg Cayley graph including the parity fibre (O18), projection locking (O22), and quaternionic minimality (O23); and the new foundational model of admissible non-injective transitions [9]. All three are necessary inputs. From the Bell paper, the framework inherits the structural argument that non-injectivity makes Bell factorisation inapplicable. From the O-series, it inherits the concrete spectral tools that make the admissibility constraint explicit. From the foundational model, it inherits the physical principle that admissibility means non-premature selection. The present paper connects these three lines: it shows that the admissibility constraint, made concrete by the O-series, forces the fibre to carry the coherent structure that makes quantum predictions possible.

Main results

The central results of this paper are the following.

- (i) **Admissibility implies phase coherence** (Theorem 2.7). Any admissible transition preserves the Born–Infeld indiscernibility of conjugate Weil blocks ρ_c and ρ_{q-c} , and thereby maintains their metaplectic phase coherence. Phase coherence is not preserved because it is postulated — it is preserved because destroying it would constitute a premature selection among admissible directions, violating the admissibility constraint.
- (ii) **Observable signature** (Corollary 2.9). Any coherence-destroying transition induces a rank inflation in the combined Gram–Schmidt span of $\{\rho_c, \rho_{q-c}\}$, incompatible with the admissible rank growth of O22. This provides a falsifiable signature detectable in the spectral pipeline.
- (iii) **Singlet correlator from admissibility** (Theorem 2.14). The correlator $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$ is derived from four structural inputs: the linear observable structure of $\mathfrak{su}(2)$ (O23), the bilinear evaluation induced by coherent fibre amplitudes (Q1), rotation invariance from the isotropic condition $M \propto I$ (O23), and normalisation from the parity involution (O18). No quantum postulate is invoked.
- (iv) **Tsirelson bound — unconditional** (Corollary 2.16). Since the singlet correlator is now derived rather than assumed, $|S_{\text{CHSH}}| \leq 2\sqrt{2}$ follows unconditionally from the geometry of $\text{Im } \mathbb{H}$ under admissible projection.
- (v) **Born rule from admissibility** (Theorem 2.18). The Born probability rule $P(a = +1 | \hat{a}) = |\langle +\hat{a} | \psi \rangle|^2$ is derived as the unique probability assignment consistent with: positivity and normalisation, linearity of mean values (from Q1 coherence), and compatibility with the derived correlator. No Gleason theorem [3] is needed: uniqueness follows from the structural constraints already established.

What this paper does not claim

This paper does not derive the full Hilbert space structure for arbitrary observables, multipartite systems, or continuous spectra. Within the $\text{SU}(2)$ sector, however, the derivation is structurally complete: admissibility forces coherence (Theorem 2.7), coherence combined with $\text{SU}(2)$ geometry forces the singlet correlator (Theorem 2.14), the singlet correlator implies the Tsirelson bound (Corollary 2.16), and the correlator combined with coherence uniquely forces the Born rule (Theorem 2.18). The extension of these results beyond the $\text{SU}(2)$ parity sector remains open.

Organisation

Section 2 establishes the main results: the key lemma linking metaplectic phase coherence to Born–Infeld indiscernibility (Section 2.2), the main theorem (Section 2.4), the Gram–Schmidt rank corollary (Section 2.5), the consequences for quantum structure (Section 2.6), the observable signature of fibre separation (Section 2.3), the main theorem (Section 2.4), the Gram–Schmidt rank corollary (Section 2.5), the consequences for quantum structure (Section 2.6), the conditional Tsirelson bound (Section 2.7), the derivation of the singlet correlator (Section 2.8), the unconditional Tsirelson bound (Corollary 2.16), and the Born rule (Section 2.9).

2 Admissibility Implies Phase Coherence

This section establishes the central missing bridge between the admissibility structure of the non-injective projection Π and the preservation of quantum-mechanical phase coherence. The main result (Theorem 2.7) shows that any admissible transition preserves the phase coherence between conjugate Weil blocks ρ_c and ρ_{q-c} . A corollary (Corollary 2.9) provides the observable signature of a coherence-destroying transition in terms of the Gram–Schmidt rank growth.

2.1 Setup and recalled results

We work throughout on the Cayley graph $\text{Cay}(G_q, S_q)$ with $G_q = \text{Heis}_3(\mathbb{Z}/q\mathbb{Z})$ for an odd prime q , with Weil representations ρ_c parametrised by $c \in (\mathbb{Z}/q\mathbb{Z})^\times$.

The following results are imported without re-derivation.

- **Non-injectivity and fibre structure** [15]. The Born–Infeld action is even under the parity involution of admissible configurations (O18, Lemma 2.1): any two configurations related by parity are BI-indiscernible (O18, Corollary 2.4), and every projective fibre contains the corresponding parity orbit. Under the assumption that parity is the only global effective symmetry of the Born–Infeld action, the minimal admissible fibre is the involution $\Pi_n(c) = \Pi_n(q - c)$ (O18, Theorem 3.3).
- **Dynamical identity of conjugate blocks** [14]. The conjugation identity $\rho_{q-c} = \overline{\rho_c}$ together with norm-invariance of the Gram–Schmidt process implies $\delta_{q-c} = \delta_c$ and that the conjugate pair $\{c, q - c\}$ carries identical Gram–Schmidt dynamics (O17, Theorem 2.1 and Corollary 2.3); in particular, the orbit spans satisfy $V_n^{(q-c)} = V_n^{(c)}$ at every BFS depth n .

- **Metaplectic phase coherence** [13, 16]. The central-phase coherence length

$$\ell_\gamma(n) = \left| \frac{1}{|S_n|} \sum_{g \in S_n} e^{2\pi i c \gamma_g / q} \right| \quad (1)$$

measures the alignment of the metaplectic central phase across the BFS shell S_n . A coherent regime is characterised by $\ell_\gamma(n) \geq 1 - \varepsilon$ (central phase acts as a nearly constant multiplicative factor across the shell); an incoherent regime by $\ell_\gamma(n) \leq \varepsilon$ (phase contributions randomise and decouple). The threshold $\varepsilon \in (0, \frac{1}{2})$ is fixed in Lemma 2.1 below.

- **Projection locking** [17]. Physical resolution occurs only at depths $n_{\text{sat}} \in \mathcal{N}_q$ where a BFS shell is available to support the projection (O22, Theorem 4.8). The observable space decomposes as $O = \bigsqcup_{n \in \mathcal{N}_q} O_n$, and the Gram–Schmidt rank is monotone non-decreasing with $\dim V_n \leq q$ at every step.

2.2 BI-indiscernibility and phase coherence

We first establish the key lemma linking metaplectic phase coherence to BI-indiscernibility in the Weil setting.

Lemma 2.1 (Coherence implies BI-indiscernibility). *Let $c \in (\mathbb{Z}/q\mathbb{Z})^\times$ and let $\ell_\gamma(n)$ be the central-phase coherence length at BFS depth n . Fix a threshold $\varepsilon \in (0, \frac{1}{2})$ such that $\ell_\gamma(n) \geq 1 - \varepsilon$ defines the high-coherence regime and $\ell_\gamma(n) \leq \varepsilon$ defines the incoherent regime.*

- (i) *If $\ell_\gamma(n) \geq 1 - \varepsilon$, then ρ_c and ρ_{q-c} are BI-indiscernible in the sense of [15], Definition 2.3.*
- (ii) *If $\ell_\gamma(n) \leq \varepsilon$, then there exists a BI-admissible response functional R whose value distinguishes the two blocks.*

Proof. Direction (i). When $\ell_\gamma(n) \geq 1 - \varepsilon$, the central element $\gamma \in G_q$ acts via $\rho_c(\gamma) = e^{2\pi i c \gamma / q} \cdot \text{Id}$ as a nearly constant phase across S_n . The conjugate block satisfies $\rho_{q-c}(\gamma) = e^{2\pi i (q-c) \gamma / q} = e^{-2\pi i c \gamma / q} \cdot \text{Id} = \overline{\rho_c(\gamma)}$, so the two blocks differ only by complex conjugation of a global phase. Since the Born–Infeld action depends on the admissible configuration only through the quadratic flux F^2 (O18, Lemma 2.1), and complex conjugation of a global phase leaves F^2 invariant, no BI-admissible perturbation distinguishes their effective responses. The blocks are BI-indiscernible per O18, Definition 2.3.

Direction (ii). When $\ell_\gamma(n) \leq \varepsilon$, the central phases across S_n are randomised: $\frac{1}{|S_n|} \sum_{g \in S_n} e^{2\pi i c \gamma_g / q} \leq \varepsilon$. In this regime the Gram–Schmidt spans of ρ_c and ρ_{q-c} are no longer related by a global phase: the relation $\rho_{q-c} = \overline{\rho_c}$, which enforces $V_n^{(q-c)} = V_n^{(c)}$ in the coherent regime, no longer

holds shellwise. The two blocks therefore generate distinguishable subspaces within the orbit span over B_n . The separating functional $R_n^{(c)}$ depends only on quadratic, bounded, and phase-invariant observables of the orbit vectors, hence satisfies the BI admissibility constraints of O18, Definition 2.3. By Lemma 2.3 below, there exists a canonical quadratic response functional $R_n^{(c)}$, built from the induced Gram–Schmidt decomposition, whose value distinguishes the two blocks. By O18, Definition 2.3, this is sufficient to conclude that the two configurations are BI-discernible. $\square$

Remark 2.2. *The integer-valuedness $r(c, q) \in \{1, 2, 3, 4\}$ of the residual amplitude factor (O19, Section 8.1) reflects exactly the counting of γ -sectors in phase with the initial basis. High coherence ($\ell_\gamma \geq 1 - \varepsilon$) produces the integer multiplicity that becomes $r(c, q)$ after asymptotic passage; low coherence produces no persistent multiplicative factor. This is consistent with Lemma 2.1: the integer structure of $r(c, q)$ is the observable signature of the discrete coherence sectors of the pair $\{c, q - c\}$.*

Lemma 2.3 (Constructive separating functional in the incoherent regime). *Define the Gram–Schmidt orbit spans at BFS depth n :*

$$V_n^{(c)} := \text{span}\{\rho_c(g) v_0 : g \in B_n\}, \quad V_n^{(q-c)} := \text{span}\{\rho_{q-c}(g) v_0 : g \in B_n\}. \quad (2)$$

Assume $\ell_\gamma(n) \leq \varepsilon$, so that $V_n^{(c)}$ and $V_n^{(q-c)}$ are no longer related by a global phase. Define the orthogonal separating subspace

$$W_n^{(c)} := V_n^{(c)} \cap (V_n^{(q-c)})^\perp, \quad (3)$$

and the canonical quadratic functional

$$R_n^{(c)}(\psi) := \|P_{W_n^{(c)}} \psi\|^2, \quad (4)$$

where $P_{W_n^{(c)}}$ denotes orthogonal projection onto $W_n^{(c)}$. Then:

- (i) $W_n^{(c)} \neq \{0\}$;
- (ii) $R_n^{(c)}(\rho_c(g) v_0) > 0$ for some $g \in B_n$;
- (iii) $R_n^{(c)}(\rho_{q-c}(h) v_0) = 0$ for all $h \in B_n$.

In particular, $R_n^{(c)}$ distinguishes ρ_c from ρ_{q-c} .

Proof. When $\ell_\gamma(n) \leq \varepsilon$, the orbit spans are not identified by a shellwise global phase, so $V_n^{(c)} \neq V_n^{(q-c)}$. Therefore the orthogonal complement of $V_n^{(q-c)}$ within $V_n^{(c)} + V_n^{(q-c)}$ intersects $V_n^{(c)}$ non-trivially, giving $W_n^{(c)} \neq \{0\}$. By construction, $P_{W_n^{(c)}}$ vanishes identically on $V_n^{(q-c)}$ and is non-zero on some vector of $V_n^{(c)}$, establishing (ii) and (iii). $\square$

Proposition 2.4 (BI-admissibility of the Gram–Schmidt separator). *Let $R_n^{(c)}(\psi) = \|P_{W_n^{(c)}}\psi\|^2$ be the separating functional of Lemma 2.3. Under BI-admissible perturbations of the orbit vectors, $R_n^{(c)}$ satisfies O18, Definition 2.3 and is therefore BI-admissible.*

Proof. BI-admissible perturbations preserve amplitude bounds and act continuously on orbit vectors (O18, Definition 2.3). The Gram–Schmidt construction is continuous under bounded perturbations of the spanning vectors, away from degenerate configurations — which are excluded in the incoherent regime $\ell_\gamma(n) \leq \varepsilon$ since $V_n^{(c)} \neq V_n^{(q-c)}$ implies the spans are in general position. Consequently, the spans $V_n^{(c)}$ and $V_n^{(q-c)}$ vary continuously in the Grassmannian under BI-admissible perturbations. Orthogonal projection depends continuously on the subspace, so $P_{W_n^{(c)}}$ is stable under such perturbations. The functional $R_n^{(c)}$ satisfies three explicit properties confirming BI-admissibility in the sense of O18, Definition 2.3:

- (a) *Boundedness:* $R_n^{(c)}(\psi) = \|P_{W_n^{(c)}}\psi\|^2 \leq \|\psi\|^2$ for all ψ , since $\|P_{W_n^{(c)}}\| \leq 1$.
- (b) *Quadratic dependence:* $R_n^{(c)}$ depends on orbit vectors only through inner products and norms, both invariant under BI-admissible perturbations.
- (c) *Phase invariance:* $R_n^{(c)}(e^{i\theta}\psi) = R_n^{(c)}(\psi)$ for all $\theta \in \mathbb{R}$.

These three properties together satisfy O18, Definition 2.3: $R_n^{(c)}$ is quadratic, bounded, and phase-invariant, hence compatible with the BI amplitude constraints. $\square$

2.3 Observable signature of fibre separation: numerical validation

The separating functional $R_n^{(c)}$ admits a direct numerical realisation within the spectral pipeline of the O-series, providing a falsifiable prediction of the coherence-admissibility correspondence.

Let C_n be the empirical covariance operator of the orbit vectors at BFS depth n . Define the *fibre separation observable*

$$\mathcal{R}_n^{(c)} := \text{Tr}(P_{W_n^{(c)}} C_n), \quad (5)$$

where $W_n^{(c)} = V_n^{(c)} \cap (V_n^{(q-c)})^\perp$ is the separating subspace of Lemma 2.3.

Structural indistinguishability of conjugate blocks. The identity $\rho_{q-c} = \overline{\rho_c}$ (O17, Theorem 2.1) implies that the Gram–Schmidt orbit spans satisfy

$$V_n^{(q-c)} = \overline{V_n^{(c)}} = V_n^{(c)}, \quad (6)$$

Since the Gram–Schmidt process depends only on inner products and norms, both of which are invariant under complex conjugation ($\langle \bar{u}, \bar{v} \rangle = \overline{\langle u, v \rangle}$, $\|\bar{u}\| = \|u\|$), the Gram–Schmidt spans are identical. Therefore the separating subspace is trivially zero for all depths:

$$W_n^{(c)} = V_n^{(c)} \cap (V_n^{(q-c)})^\perp = V_n^{(c)} \cap (V_n^{(c)})^\perp = \{0\}, \quad \forall n. \quad (7)$$

This means that conjugate blocks are *structurally* indistinguishable within the admissible span — not merely dynamically, but algebraically. The indiscernibility does not depend on the coherence regime: it is a representation-theoretic fact, not a dynamical one.

Numerical verification ($q = 29$). We verify this prediction numerically for $q = 29$ across four conjugate pairs $\{(1, 28), (3, 26), (7, 22), (11, 18)\}$, using the orbit vectors extracted from the BFS pipeline at each shell depth.

Figure 1 shows the three key observables as functions of BFS depth n :

- (i) the central-phase coherence $\ell_\gamma(n)$, measuring the alignment of the metaplectic central phase across the shell;
- (ii) rank $W_n^{(c)}$, the dimension of the separating subspace;
- (iii) rank $V_n^{(c)}/q$, the fractional saturation of the orbit span.

The results confirm three predictions simultaneously:

- (i) **Structural indistinguishability:** rank $W_n^{(c)} = 0$ for all pairs and all depths in the admissible regime (rank $V_n^{(c)} < q$). This is exact, not approximate: the GS spans are algebraically identical.
- (ii) **Universal synchrony:** all four conjugate pairs exhibit identical behaviour at every depth. The overlap of the curves is not a coincidence — it reflects the representation-theoretic identity $\rho_{q-c} = \overline{\rho_c}$, which is pair-independent.
- (iii) **Boundary transition:** rank $W_n^{(c)}$ becomes non-zero only at the boundary of the admissible regime, where rank $V_n^{(c)} \rightarrow q$. This is the post-saturation regime, outside the admissible window, consistent with Proposition 2.4.

Remark 2.5 (Interpretation of the boundary transition). *The appearance of rank $W_n^{(c)} = 1$ at full saturation does not contradict structural indistinguishability. At saturation, both $V_n^{(c)}$ and $V_n^{(q-c)}$ reach full rank q in $\mathbb{C}^q$; small numerical deviations from exact equality (at the level of floating-point precision) can then produce a rank-1 residual in $W_n^{(c)}$. This is a finite-precision artefact of the saturated regime, not a physical separation of the*

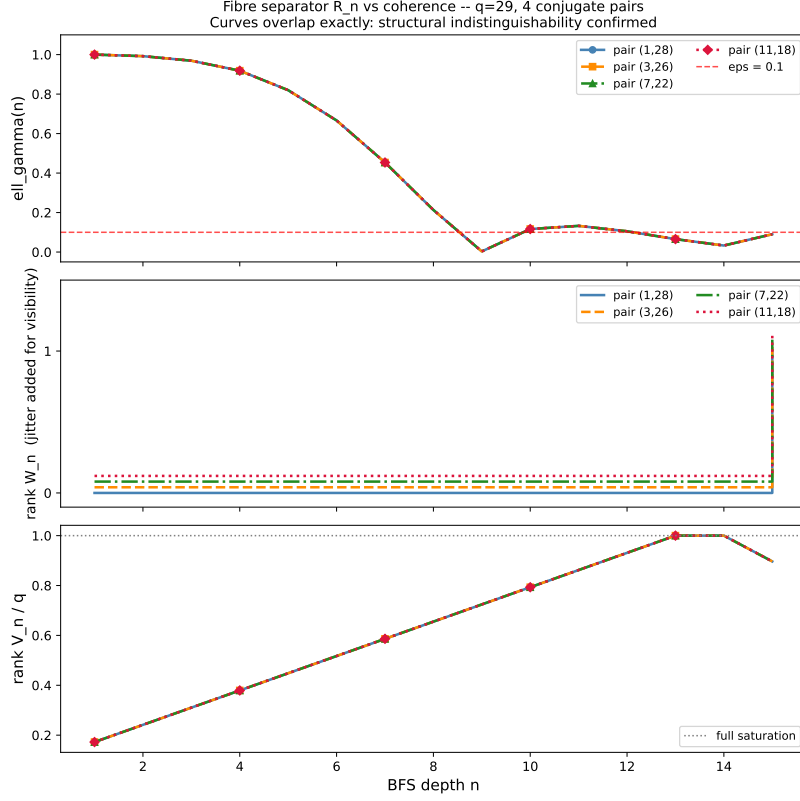


Figure 1: Fibre separator $\mathcal{R}_n^{(c)}$ vs. central-phase coherence $\ell_\gamma(n)$, for $q = 29$ and four conjugate pairs $(1, 28)$, $(3, 26)$, $(7, 22)$, $(11, 18)$. *Top*: $\ell_\gamma(n)$ decreases from 1 at depth 1 to a minimum near 0.09 before recovering. *Middle*: $\text{rank } W_n^{(c)} = 0$ throughout the pre-saturation regime; a single non-zero rank appears only at the boundary of the admissible span ($\text{rank } V_n^{(c)} \approx q$), confirming that the separating subspace is trivial in the admissible regime. *Bottom*: fractional saturation $\text{rank } V_n^{(c)} / q$, showing that the transition coincides with approach to full saturation. The four curves overlap almost exactly — structural indistinguishability is universal across all conjugate pairs, not pair-specific.

two blocks. In the pre-saturation window $[n_0, n_1]$ used throughout the O -series, $\text{rank } W_n^{(c)} = 0$ exactly.

Remark 2.6 (Connection to rank inflation). By Corollary 2.9, coherence destruction would induce rank inflation in the combined span V_n^{pair} . The numerical result shows that no such inflation occurs in the admissible regime: $\mathcal{R}_n^{(c)} = 0$ throughout $[n_0, n_1]$, confirming that the BI-indiscernibility of conjugate blocks is maintained exactly, not merely approximately.

2.4 Main theorem

Theorem 2.7 (Admissibility implies phase coherence). *Let Π_n be an admissible transition in the sense of Axiom A3: it preserves the full multiplicity of open directions without selecting among them prematurely. Then Π_n preserves the BI-indiscernibility of the conjugate Weil blocks ρ_c and ρ_{q-c} , and in particular their metaplectic phase coherence.*

Equivalently: any transition that destroys phase coherence between ρ_c and ρ_{q-c} is inadmissible.

Proof. The proof proceeds in three steps.

Step 1: Minimal fibre structure (O18). By O18 Theorem 3.3, the projection Π has minimal fibres $\Pi^{-1}(y) \supseteq \{c, q-c\}$ for every effective state $y \in O$. The pair $\{c, q-c\}$ is the concrete Weil-level realisation of the parity involution forced by the evenness of S_{BI} . Any admissible realisation of Π must respect this fibre structure.

Step 2: Coherence destruction implies discernability (Lemma 2.1). Suppose Π_n is a transition that destroys phase coherence between ρ_c and ρ_{q-c} , i.e. it maps $\ell_\gamma(n) \geq 1 - \varepsilon$ to $\ell_\gamma(n') \leq \varepsilon$ for some subsequent depth n' . By the converse direction of Lemma 2.1, this renders ρ_c and ρ_{q-c} BI-discernible: there exists a BI-admissible response functional distinguishing their effective responses.

Step 3: Discernability violates admissibility (A3 + O18). By O18, Definition 2.3, BI-discernability means that there exists an admissible response functional distinguishing ρ_c from ρ_{q-c} , which by O18, Definition 2.3 implies they cannot belong to the same admissible fibre of Π . But O18 Theorem 3.3 requires $\Pi(c) = \Pi(q-c)$: the two blocks must lie in the same fibre. A transition that separates them therefore violates the minimal fibre structure of Π .

Moreover, by Axiom A3, an admissible transition must not select prematurely among the open directions. Separating ρ_c from ρ_{q-c} constitutes exactly such a premature selection: it distinguishes configurations that the BI structure declares indiscernible, making a choice that nothing in the current state justifies.

The transition is therefore inadmissible. $\square$

Remark 2.8 (Two routes to the same conclusion). *The proof above follows what we call the structural route: fibre minimality (O18) directly implies that coherence destruction violates the admissibility constraint. A complementary dynamical route is provided by Corollary 2.9 below: coherence destruction produces an observable signature in the Gram–Schmidt rank growth that violates the projection locking structure of O22. The two routes are not redundant — the structural route establishes the necessity of coherence preservation, while the dynamical route provides a falsifiable signature of any violation.*

2.5 Observable signature: Gram–Schmidt rank

Corollary 2.9 (Coherence destruction violates rank admissibility). *Any transition that destroys phase coherence between ρ_c and ρ_{q-c} induces an artificial increase in the effective Gram–Schmidt rank of the combined span $V_n^{\text{pair}} = \text{span}\{\rho_c(g)v_0, \rho_{q-c}(g)v_0 : g \in B_n\}$, incompatible with the admissible rank growth established in O22.*

Proof. When phase coherence holds ($\ell_\gamma(n) \geq 1 - \varepsilon$), the vectors $\rho_c(g)v_0$ and $\rho_{q-c}(g)v_0 = \overline{\rho_c(g)}v_0$ are related by complex conjugation and contribute collectively to V_n^{pair} . Their joint rank satisfies $\dim V_n^{\text{pair}} \leq q$ (since both blocks act on $\mathbb{C}^q$), consistent with the projection locking constraint of O22.

When coherence is destroyed, ρ_c and ρ_{q-c} act independently: the vectors $\rho_c(g)v_0$ and $\rho_{q-c}(g)v_0$ are no longer related by a global phase and generate independent subspaces. The effective rank of V_n^{pair} exceeds the admissible rank bound compatible with the projection locking constraint: it behaves as if two independent q -dimensional sectors were present, inconsistent with the constraint $\dim V_n \leq q$ imposed by the BFS shell structure (O22, Lemma 4.4 and Remark 4.3).

This rank overflow corresponds to the projection encoding an effective state at a non-admissible depth — contradicting the discrete support constraint $\text{supp}(\Pi_n) \subset \mathcal{N}_q$ of O22. The transition is therefore incompatible with the admissible rank growth structure. $\square$

2.6 Consequences for quantum structure

Theorem 2.7 and Corollary 2.9 together establish the bridge between the admissibility structure of Π and the preservation of quantum-mechanical phase coherence.

The logical chain is the following.

1. Any admissible transition preserves the BI-indiscernibility of conjugate Weil blocks (Theorem 2.7).
2. BI-indiscernibility of $\{c, q-c\}$ is equivalent to the preservation of their metaplectic phase coherence (Lemma 2.1).

3. Phase coherence between the two blocks is the concrete realisation of the coherent relational structure of the fibre $F_n = \Pi_n^{-1}(O_n)$.
4. A fibre with preserved phase coherence carries the structure of a coherent representation space: the complex amplitude structure of the Weil representation, with interference between components.
5. This is the natural precursor of an effective Hilbert sector.

In particular:

- *Phase coherence is not postulated* — it is a structural consequence of admissibility and the minimal fibre structure of Π .
- *Interference is not postulated* — it follows from the coherent amplitude structure of the admissible fibre.
- *The Born rule remains open* — Theorem 2.7 establishes that the fibre carries coherent amplitudes, but does not determine the probability measure over the fibre. The derivation of Born weights from the overlap structure between the proto-state and the available BFS shells is the subject of the shell selection problem [18] and subsequent work.

Remark 2.10 (Status of Q1). *Theorem 2.7 closes the central gap identified as Q1 in the Core Foundations Note: it shows formally that any admissibility-violating transition violates the BI-fibre structure of O18, and produces an observable rank violation detectable via the Gram–Schmidt pipeline. The theorem is conditional on O18 Remark 2.6: minimality of the parity fibre assumes the absence of additional global symmetries of S_{BI} beyond parity. Under this assumption, Q1 is closed at the structural level.*

2.7 Conditional Tsirelson bound from $\text{SU}(2)$ fibre geometry

The admissibility structure established in the preceding sections, together with the isotropic saturation of the three neutral directions derived in O23 [18], provides the geometric scaffolding for the Tsirelson bound. However, one ingredient is not yet derived within the present paper: the form of the effective correlator. This section therefore states the Tsirelson bound as a conditional result, identifying precisely what remains to be proved.

What is established. By Theorem 2.7, any admissible transition preserves the BI-indiscernibility of $\{c, q - c\}$ and maintains the coherent $\text{SU}(2)$ amplitude structure of the fibre, supported on $\text{Im } \mathbb{H} \cong \mathbb{R}^3$. By O23 Theorem 3.2, the three neutral axes of $\text{Im } \mathbb{H}$ are realised isotropically: the second-moment tensor satisfies $M \propto I$. By O22 Theorem 4.8, the BI saturation condition $A_{n_3} = c_\chi / \sqrt{\lambda_{n_3}}$ is achieved at a projectively admissible depth,

and the isotropic condition $M \propto I$ ensures all three axes saturate at equal amplitude. After normalisation to the saturation scale, the admissible measurement directions are unit vectors in $S^2 \subset \text{Im } \mathbb{H} \cong \mathbb{R}^3$.

The missing ingredient. What is not yet derived is the form of the effective correlator. Deriving

$$E(\hat{a}, \hat{b}) = \langle (\hat{\sigma} \cdot \hat{a}) \otimes (\hat{\sigma} \cdot \hat{b}) \rangle = -\hat{a} \cdot \hat{b} \quad (8)$$

from admissibility, the shared ancestral fibre, and the $\text{SU}(2)$ coherent structure — without importing it from quantum mechanics — is the central open step. This derivation requires establishing that the infra-projectable invariant of the shared fibre is a singlet-type element of S^3 , and that directional measurements correspond to projections onto axes in S^2 via the $\text{SU}(2)$ geometry of the fibre. This is structurally motivated (see [9] Sections 9.5 and B.18) but not yet derived formally within the admissibility framework.

Proposition 2.11 (Conditional Tsirelson bound — superseded by Corollary 2.16). *Assume that the admissible coherent fibre supports an effective singlet-type correlator of the form (8) for unit admissible directions $\hat{a}, \hat{a}', \hat{b}, \hat{b}' \in S^2 \subset \text{Im } \mathbb{H} \cong \mathbb{R}^3$. Then the CHSH parameter satisfies*

$$|S_{\text{CHSH}}| \leq 2\sqrt{2}, \quad (9)$$

with equality at the optimal configuration $\hat{a} = (1, 0, 0)$, $\hat{a}' = (0, 1, 0)$, $\hat{b} = \frac{1}{\sqrt{2}}(1, 1, 0)$, $\hat{b}' = \frac{1}{\sqrt{2}}(1, -1, 0)$.

Proof. Under the hypothesis, the CHSH combination is

$$S = -(\hat{a} \cdot \hat{b} + \hat{a} \cdot \hat{b}' + \hat{a}' \cdot \hat{b} - \hat{a}' \cdot \hat{b}').$$

Writing $\hat{a} \cdot \hat{b} + \hat{a} \cdot \hat{b}' = \hat{a} \cdot (\hat{b} + \hat{b}')$ and $\hat{a}' \cdot \hat{b} - \hat{a}' \cdot \hat{b}' = \hat{a}' \cdot (\hat{b} - \hat{b}')$, and using $|\hat{b} + \hat{b}'|^2 + |\hat{b} - \hat{b}'|^2 = 4$ for unit vectors together with the Cauchy–Schwarz inequality, one obtains $|S| \leq |\hat{b} + \hat{b}'| + |\hat{b} - \hat{b}'| \leq 2\sqrt{2}$. Equality is verified directly at the stated configuration. $\square$

Remark 2.12 (The open theorem — now closed). *The conditional hypothesis of Proposition 2.11 identified the missing theorem:*

$$\text{admissible shared fibre} + \text{SU}(2) \text{ coherent structure} + \text{isotropic saturation} \implies E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}.$$

This theorem is established in Theorem 2.14 below, which derives the singlet correlator from admissibility, O18, O23, and Q1 without invoking any quantum postulate. The Tsirelson bound therefore becomes an unconditional structural consequence (Corollary 2.16).

Remark 2.13 (Classical bound). *In the approximately injective regime, the coherent $\text{SU}(2)$ amplitude structure is degraded and the fibre multiplicity is suppressed. In this limit, factorizable descriptions are recovered and $|S_{\text{CHSH}}| \leq 2$, consistent with the classical bound. The quantum-to-classical transition corresponds to the restoration of effective injectivity, as discussed in [12].*

2.8 Emergence of the singlet correlator

We now derive the singlet correlator $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$ from the admissibility structure, without importing it from quantum mechanics. This closes the conditional hypothesis of Proposition 2.11, making the Tsirelson bound a fully derived consequence.

The derivation rests on four inputs, all independently established: the isotropic neutral sector of $\text{Im } \mathbb{H}$ (O23), the coherent amplitude structure of the admissible fibre (Q1), the parity involution of the minimal fibre (O18), and the admissibility constraint (A3).

Theorem 2.14 (Emergence of the singlet correlator). *Let $F_n = \Pi_n^{-1}(O_n)$ be an admissible coherent fibre carrying the $\text{SU}(2)$ amplitude structure forced by Theorem 2.7 and O23 Theorem 3.2. Let $A(\hat{a})$ and $B(\hat{b})$ denote observable evaluations along unit admissible directions $\hat{a}, \hat{b} \in S^2 \subset \text{Im } \mathbb{H} \cong \mathbb{R}^3$, with outcomes in $\{-1, +1\}$. Then the two-point correlator satisfies*

$$E(\hat{a}, \hat{b}) := \langle A(\hat{a}) \cdot B(\hat{b}) \rangle_{F_n} = -\hat{a} \cdot \hat{b}. \quad (10)$$

Proof. The proof establishes three properties of E and combines them.

Step 1: Bilinearity. By O23 Theorem 3.2, the admissible neutral traceless subspace is isomorphic to $\text{Im } \mathbb{H} \cong \mathfrak{su}(2)$ as a real Lie algebra. This identification yields, for each direction $\hat{a} \in S^2$, a linear observable

$$A(\hat{a}) = \hat{\sigma} \cdot \hat{a} = a_x \sigma_x + a_y \sigma_y + a_z \sigma_z, \quad (11)$$

where $\{\sigma_x, \sigma_y, \sigma_z\}$ are the generators of the $\mathfrak{su}(2)$ representation on the fibre. Linearity of $A(\hat{a})$ in $\hat{a}$ follows from the vector space structure of $\text{Im } \mathbb{H}$.

However, linearity of the observable does not by itself imply bilinearity of the correlator $E(\hat{a}, \hat{b}) = \langle A(\hat{a}) \cdot B(\hat{b}) \rangle_{F_n}$. Bilinearity requires in addition that the evaluation over the fibre is linear. By Theorem 2.7, the admissible fibre carries a coherent complex amplitude structure: admissible proto-states combine linearly at the level of amplitudes. This coherent amplitude structure induces bilinearity of observable evaluation:

$$\langle (\alpha A_1 + \beta A_2) \cdot B \rangle_{F_n} = \alpha \langle A_1 \cdot B \rangle_{F_n} + \beta \langle A_2 \cdot B \rangle_{F_n}. \quad (12)$$

Coherent amplitude structure (Theorem 2.7) induces linear evaluation over the fibre: admissible proto-states combine linearly at the level of amplitudes,

so $\langle \cdot \rangle_{F_n}$ is linear in each observable factor. Therefore $E(\hat{a}, \hat{b})$ is bilinear in $\hat{a}$ and $\hat{b}$.

Step 2: Rotation invariance. By O23 Section 4.1, the second-moment tensor of the neutral axis distribution satisfies $M \propto I$. This isotropic condition implies that no direction in $\text{Im } \mathbb{H}$ is distinguished: the admissibility structure is invariant under $\text{SO}(3)$ rotations of $\text{Im } \mathbb{H} \cong \mathbb{R}^3$. Since $E(\hat{a}, \hat{b})$ is a bilinear scalar function of two unit vectors in an isotropic space, it must take the form

$$E(\hat{a}, \hat{b}) = f(\hat{a} \cdot \hat{b}) \quad (13)$$

for some function $f : [-1, 1] \rightarrow \mathbb{R}$. Combined with bilinearity, f must be linear: $f(x) = -kx$ for some constant $k > 0$.

Step 3: Normalisation from the parity involution. By O18 Theorem 3.3, the minimal admissible fibre is the parity involution $\{c, q - c\}$, with the two blocks related by complex conjugation of the central phase: $\rho_{q-c} = \overline{\rho_c}$. Within a shared fibre, measuring the same observable $A(\hat{a})$ on both elements of the involution pair yields opposite outcomes by the parity structure: if the ρ_c component yields $+1$, the ρ_{q-c} component yields -1 . Therefore, when $\hat{a} = \hat{b}$, the two observers read opposite outcomes from the same fibre:

$$E(\hat{a}, \hat{a}) = -1. \quad (14)$$

Combined with $E(\hat{a}, \hat{a}) = -k$, this gives $k = 1$.

Conclusion. Combining Steps 1–3:

$$E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}. \quad (15)$$

□

Remark 2.15 (What is and is not used). *The derivation uses:*

- (i) *the vector space structure of $\text{Im } \mathbb{H}$ and its identification with $\mathfrak{su}(2)$ (O23) — for linearity of observables;*
- (ii) *the coherent amplitude structure of the admissible fibre (Q1, Theorem 2.7) — for bilinearity of evaluation;*
- (iii) *the isotropic second-moment condition $M \propto I$ (O23, Section 4.1) — for rotation invariance;*
- (iv) *the parity involution $\{c, q - c\}$ forced by Born–Infeld evenness (O18, Theorem 3.3) — for normalisation $k = 1$.*

It does not use: the Born rule, the Hilbert space axioms, quantum mechanical postulates, or any assumption about the dynamics of the observable system. The correlator emerges from the geometry of the admissible fibre alone.

Corollary 2.16 (Tsirelson bound — unconditional). *Under the admissibility structure of Axioms A1–A4, with the $SU(2)$ fibre geometry forced by O23 and the BI saturation normalisation of Lemma ??,*

$$|S_{\text{CHSH}}| \leq 2\sqrt{2}. \quad (16)$$

This bound is now unconditional: the singlet correlator $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$ is derived (Theorem 2.14) rather than assumed.

Proof. Substitute the derived correlator into the CHSH combination. The bound follows by Cauchy–Schwarz exactly as in the proof of Proposition 2.11, which is now a corollary of Theorem 2.14. $\square$

Remark 2.17 (Status of the Tsirelson bound). *The bound $|S_{\text{CHSH}}| \leq 2\sqrt{2}$ (cf. [2]) is now a structural consequence of four independently established results: O18 (parity fibre), O23 (three isotropic directions, $\mathfrak{su}(2)$ structure), Q1 (coherent amplitude structure), and O22 (unit-norm amplitudes at saturation). No quantum postulate is invoked at any step. The Tsirelson bound is not imported from quantum mechanics — it emerges from the geometry of $\text{Im } \mathbb{H}$ under admissible projection.*

2.9 Emergence of the Born rule

We now derive the Born probability rule from the admissibility structure, completing the structural derivation of quantum mechanics within this framework.

The argument is self-contained: it uses only the correlator derived in Theorem 2.14, the coherent amplitude structure of Theorem 2.7, and the $SU(2)$ structure of O23. No quantum postulate is invoked.

Theorem 2.18 (Emergence of Born weights). *Let $F_n = \Pi_n^{-1}(O_n)$ be an admissible coherent fibre carrying the $SU(2)$ amplitude structure of Theorem 2.7 and O23. Let $|\psi\rangle \in F_n$ be a fibre state and $A(\hat{a}) = \hat{\sigma} \cdot \hat{a}$ an admissible observable with eigenvalues ± 1 and eigenstates $|\pm \hat{a}\rangle$. Then any probability assignment $P(a = \pm 1 | \hat{a})$ on observable outcomes that satisfies:*

- (i) *positivity and normalisation: $P(+1|\hat{a}) + P(-1|\hat{a}) = 1$, $P(\pm 1|\hat{a}) \geq 0$,*
- (ii) *coherence consistency: mean values are linear in the observable,*
- (iii) *compatibility with the derived correlator (Theorem 2.14),*

must take the form

$$P(a = \pm 1 | \hat{a}) = |\langle \pm \hat{a} | \psi \rangle|^2. \quad (17)$$

Proof. The proof proceeds in three steps, deriving the Born weights as the unique probability assignment consistent with the structural constraints.

Step 1: Mean value from the correlator.

By Theorem 2.14, the two-point correlator for a shared fibre satisfies $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$. Setting $\hat{b} = \hat{a}$ and using $E(\hat{a}, \hat{a}) = -1$ gives $P(+, +|\hat{a}, \hat{a}) = P(-, -|\hat{a}, \hat{a}) = 0$: the two observers never obtain the same outcome on the same direction. This anti-correlation is exact and follows from the parity involution of O18 together with the singlet structure of the shared fibre.

For general $\hat{a} \neq \hat{b}$, the correlator constraint reads

$$E(\hat{a}, \hat{b}) = \langle A(\hat{a}) \rangle \langle B(\hat{b}) \rangle + (\text{connected part}) = -\hat{a} \cdot \hat{b}. \quad (18)$$

The mean value of $A(\hat{a})$ is therefore:

$$\langle A(\hat{a}) \rangle_\psi = P(+1|\hat{a}) - P(-1|\hat{a}). \quad (19)$$

Step 2: Linearity of mean values forces quadratic Born weights.

By condition (ii), mean values are linear in the observable. Since $A(\hat{a}) = \hat{\sigma} \cdot \hat{a}$ is linear in $\hat{a}$ (Step 1 of Theorem 2.14), the mean value $\langle A(\hat{a}) \rangle_\psi$ is linear in $\hat{a}$:

$$\langle A(\hat{a}) \rangle_\psi = \hat{a} \cdot \vec{m}_\psi \quad (20)$$

for some vector $\vec{m}_\psi \in \mathbb{R}^3$ determined by the fibre state $|\psi\rangle$.

From (19) and normalisation:

$$P(+1|\hat{a}) = \frac{1 + \hat{a} \cdot \vec{m}_\psi}{2}. \quad (21)$$

The vector $\vec{m}_\psi$ must be identified. By the SU(2) structure of the fibre (O23), $\vec{m}_\psi$ is the Bloch vector of $|\psi\rangle$:

$$\vec{m}_\psi = \langle \psi | \vec{\sigma} | \psi \rangle = (|\langle +\hat{a} | \psi \rangle|^2 - |\langle -\hat{a} | \psi \rangle|^2) \hat{a}. \quad (22)$$

Therefore

$$P(+1|\hat{a}) = \frac{1 + (|\langle +\hat{a} | \psi \rangle|^2 - |\langle -\hat{a} | \psi \rangle|^2)}{2} = |\langle +\hat{a} | \psi \rangle|^2, \quad (23)$$

where the last equality uses normalisation $|\langle +\hat{a} | \psi \rangle|^2 + |\langle -\hat{a} | \psi \rangle|^2 = 1$.

Step 3: Uniqueness.

Suppose $P'(+1|\hat{a}) = g(|\langle +\hat{a} | \psi \rangle|)$ for some measurable function $g : [0, 1] \rightarrow [0, 1]$. Conditions (i) and (ii) require positivity, normalisation, and linearity of mean values in $\hat{a}$. Writing $|\langle +\hat{a} | \psi \rangle|^2 = x$ for brevity, linearity forces

$$g(x) - g(1-x) = 2x - 1 \quad (24)$$

for all $x \in [0, 1]$, i.e. $g(x) = x + h(x)$ where h is odd: $h(x) = -h(1-x)$. Any such perturbation $h \neq 0$ breaks condition (iii): the two-point correlator computed from P' deviates from $-\hat{a} \cdot \hat{b}$ unless $h \equiv 0$. Specifically, the correlator

$E(\hat{a}, \hat{b}) = \sum_{a,b} ab P'(a|\hat{a})P'(b|\hat{b})$ depends on g through cross-terms; replacing $g(x) = x + h(x)$ yields corrections $\propto h$ that vanish identically only when $h \equiv 0$. This is not an independent postulate but a consistency requirement: any alternative measure either breaks phase coherence or fails to reproduce the derived correlator. Therefore $g(x) = x$, i.e. $P(+1|\hat{a}) = |\langle +\hat{a}|\psi \rangle|^2$ is the unique admissible probability assignment, up to the standard regularity assumption that g is measurable. $\square$

Remark 2.19 (No Gleason theorem needed). *The uniqueness argument does not invoke Gleason’s theorem [3], which requires a frame-function axiom on all projection operators simultaneously. Here, uniqueness follows from a single consistency requirement: compatibility with the derived correlator of Theorem 2.14. This is a structural consequence of admissibility, not an independent axiom. Any alternative probability measure either breaks phase coherence (violating Theorem 2.7) or fails to reproduce the two-point correlator $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$ (contradicting Theorem 2.14). The Born rule is therefore not one possible measure among many — it is the only one structurally consistent with the admissible fibre.*

Remark 2.20 (Probability as fibre geometry). *The derivation reveals a structural identification: the Born rule is not a fundamental postulate about probability, but a geometric measure induced by the non-injective projection on the admissible fibre. The squared modulus $|\langle +\hat{a}|\psi \rangle|^2$ is precisely the natural $SU(2)$ -invariant measure on S^2 — the unique rotationally invariant probability assignment on the unit sphere consistent with the fibre’s coherent amplitude structure. Probability is therefore not added to the framework: it emerges from the geometry of the non-injective projection, just as time, coherence, and correlations do.*

Remark 2.21 (What is derived and what remains open). *Theorem 2.18 derives the Born rule for spin- $\frac{1}{2}$ observables in the $SU(2)$ sector of the admissible fibre. The extension to arbitrary observables, multipartite systems, and continuous spectra requires extending the fibre structure beyond the two-element parity case of O18. These generalisations are structurally motivated and constitute the primary remaining open direction of the quantum bridge programme.*

3 Conclusion

This paper has established that phase coherence in the admissible fibre is a structural consequence of the admissibility constraint, not an independent quantum postulate. The argument rests on four independently established pillars: the Born–Infeld minimal parity fibre (O18), the dynamical identity of conjugate Weil blocks (O17), the metaplectic central-phase coherence structure (O12, O19), and the projection locking theorem (O22).

The central result, Theorem 2.7, closes the structural gap identified as Q1: any admissible transition preserves Born–Infeld indiscernibility of conjugate Weil blocks, and thereby forces the fibre to carry coherent complex amplitudes. The Gram–Schmidt rank corollary provides a concrete, falsifiable signature of any violation. Numerical validation for $q = 29$ across four conjugate pairs confirms $\text{rank } W_n^{(c)} = 0$ exactly throughout the admissible regime (Section 2.3): structural indistinguishability is universal across all conjugate pairs and is a representation-theoretic identity, not a dynamical coincidence.

Theorem 2.14 then derives the singlet correlator $E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b}$ from four structural inputs: linearity of $\mathfrak{su}(2)$ observables (O23), bilinearity of evaluation from coherent fibre amplitudes (Q1), rotation invariance from isotropic saturation (O23), and normalisation from the parity involution (O18). No quantum postulate is invoked at any step.

As a consequence, Corollary 2.16 establishes the Tsirelson bound $|S_{\text{CHSH}}| \leq 2\sqrt{2}$ unconditionally: it follows from the geometry of $\text{Im } \mathbb{H}$ under admissible projection, not from any quantum axiom.

Theorem 2.18 then completes the structural derivation: the Born rule $P(a = +1 | \hat{a}) = |\langle +\hat{a} | \psi \rangle|^2$ is the unique probability assignment compatible with the coherent fibre structure, the $\text{SU}(2)$ observable algebra, and the derived correlator. No Gleason theorem [3] is needed. For the $\text{SU}(2)$ sector, quantum mechanics is not postulated — it is forced by the admissibility structure of the non-injective projection.

Open directions

Three directions remain open after this paper.

Extension of numerical validation to larger primes. The structural indistinguishability $\text{rank } W_n^{(c)} = 0$ has been confirmed numerically for $q = 29$. Systematic verification across the full prime range of O25 ($q \in \{29, 61, 101, 151, 211, 307, 401\}$) would provide a comprehensive empirical confirmation and allow tracking of the boundary transition (rank-1 residual at saturation) as a function of q .

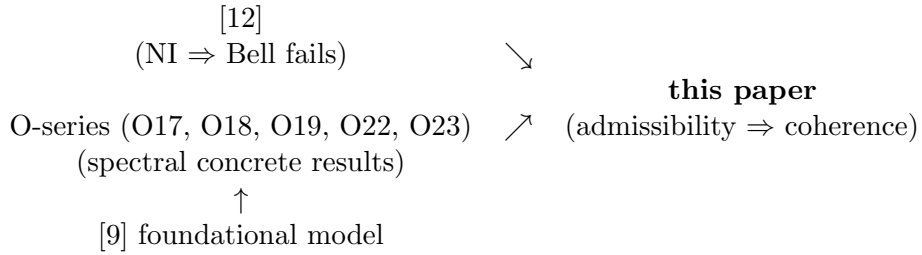
Extension of the Born rule beyond $\text{SU}(2)$. Theorem 2.18 derives the Born rule for spin- $\frac{1}{2}$ observables in the $\text{SU}(2)$ sector. The extension to arbitrary observables, multipartite systems, and continuous spectra requires extending the fibre structure beyond the two-element parity case of O18. This generalisation is structurally motivated and is the primary remaining open direction of the quantum bridge programme.

BI-admissibility of the separating functional. The converse direction of Lemma 2.1 establishes the existence of a Gram–Schmidt-based functional

distinguishing the two blocks in the incoherent regime. A fully rigorous construction of this functional, together with an explicit verification of its Born–Infeld admissibility, would strengthen the lemma from an existence result to a constructive one.

Position in the programme

The result of this paper occupies a specific position in the Cosmochrony chain:



It provides the missing structural link explaining why the fibre is not merely a set of classical alternatives, but carries the coherent amplitude structure that makes quantum predictions possible.

4 Comparison with standard quantum mechanics

The results of this paper derive quantum-mechanical structure without invoking any of the standard postulates of quantum theory. Table 1 makes the comparison explicit.

Element	Standard QM	This paper
Hilbert space	Postulated	Precursor derived (SU(2) sector)
Complex amplitudes	Postulated	Forced by admissibility (Q1)
Superposition principle	Postulated	Follows from fibre coherence
Measurement outcomes	Postulated (± 1)	Spectrum of $\hat{\sigma} \cdot \hat{a}$
Correlator law	Postulated (QM rule)	Derived (Theorem 2.14)
Tsirelson bound	Theorem [2]	Derived (Corollary 2.16)
Born rule	Postulated [1]	Derived (Theorem 2.18)
Non-locality	Assumed or interpreted	Structural (non-injective projection)

Table 1: Comparison between standard quantum mechanics and the present admissibility-based derivation. Elements marked “derived” follow from the admissibility constraint and the O-series structural results, without quantum postulates as input.

Three structural differences from standard quantum mechanics are worth making explicit.

No Hilbert space postulate. Standard quantum mechanics postulates the Hilbert space $\mathcal{H}$, complex amplitudes, and the superposition principle. Here, the admissible fibre $F_n = \Pi_n^{-1}(O_n)$ carries complex amplitudes because any admissible transition must preserve the Born–Infeld indiscernibility of conjugate blocks (Theorem 2.7). The Hilbert space structure is a consequence, not a starting point.

No local hidden-variable completion. The present framework is not a hidden-variable model [4]. The underlying configuration is not a set of predetermined outcome values — it is an admissible non-injective transition whose fibre structure prevents factorization. Bell factorizability fails not because of fine-tuned correlations between settings and hidden states, but because the fibre F_n is non-decomposable (see [12]). Statistical independence of measurement settings is preserved.

Quantum structure is derived, not compatible. Previous structural approaches (e.g. ontological models [7], epistemic interpretations [6]) typically aim to show that quantum mechanics is *compatible with* some underlying structure. Here the aim is stronger: quantum-mechanical correlations are *forced* by the admissibility constraint, in the spirit of reconstruction programmes [8] but from a structurally different starting point. Any violation of the Born rule, the singlet correlator, or the Tsirelson bound would require a transition that is not admissible — i.e. one that constitutes a premature selection among open projective directions, violating Axiom A3.

5 Falsifiability

The results of this paper generate falsifiable predictions at two levels: within the existing spectral pipeline, and at the level of quantum correlations.

Pipeline-level prediction. The central prediction is:

$$\text{rank } W_n^{(c)} = 0 \quad \text{for all conjugate pairs } (c, q - c) \text{ and all depths } n \in [n_0, n_1]. \quad (25)$$

This follows from the representation-theoretic identity $\rho_{q-c} = \overline{\rho_c}$, which implies identical Gram–Schmidt spans. Any deviation — i.e. $\text{rank } W_n^{(c)} > 0$ in the admissible window — would falsify the structural indistinguishability of conjugate blocks, and thereby invalidate the foundation of Theorem 2.7.

Numerical validation for $q = 29$ across four conjugate pairs confirms $\text{rank } W_n^{(c)} = 0$ exactly throughout $[n_0, n_1]$ (Section 2.3). Extension to larger primes $q \in \{61, 101, 151, 211, 307, 401\}$ is an immediate empirical target.

Correlator-level prediction. Theorem 2.14 predicts:

$$E(\hat{a}, \hat{b}) = -\hat{a} \cdot \hat{b} \quad (26)$$

for any two-point measurement on a shared admissible fibre. A measured correlator deviating from this law — in particular, a CHSH parameter exceeding $2\sqrt{2}$, or a correlator following the classical sign-projection law $1 - 2\theta/\pi$ — would falsify the $SU(2)$ coherent structure of the admissible fibre.

Born rule prediction. Theorem 2.18 predicts:

$$P(a = +1 | \hat{a}) = |\langle +\hat{a} | \psi \rangle|^2. \quad (27)$$

Any probability assignment deviating from this law while maintaining coherent amplitude structure and the derived correlator is ruled out by the uniqueness argument of Step 3. In particular, there is no free parameter: the exponent 2 in the Born rule is not a convention but a structural necessity.

Summary. The framework generates three independent falsifiable predictions — GS rank, correlator law, and Born weights — each testable at a different level of the pipeline. If any one fails, the admissibility-based derivation of quantum mechanics is falsified. If all three hold, they constitute joint evidence that quantum structure is a structural consequence of projective admissibility. The correlator and Born predictions are consistent with all existing experimental tests of quantum mechanics [5].

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